



The Effects of Partners' Helping Motivation on Chronic Pain Patients' Functioning Over Time

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Abstract: To understand when and why the provision of help by a partner of an individual with chronic pain (ICP) yields benefits, it is critical, according to self-determination theory, to consider the extent to which partners' helping responses are supportive of the basic psychological needs of the ICP, as well as the motivations underlying these helping responses. The present study (N = 141 couples), spanning 3 measurement moments over 6 months, investigated temporal associations between partners' helping motivation, ICPs' psychological needs, and ICPs' functioning across time (ie, well-being, psychological distress, and disability). Results showed that partners' autonomous or volitional helping motivation (time 1) predicted decreases in ICPs' need frustration (time 2) and ICPs' need frustration (time 2) predicted increases in ICPs' psychological distress (time 3). Further, ICPs' need satisfaction (time 2) predicted increases in well-being (time 3) and decreases in psychological distress (time 3). The link between need frustration and ICPs' well-being (time 1-time 2) was bidirectional, with both relating reciprocally to one another over time. Finally, the associations between ICPs' disability and both partners' helping motivation and ICPs' need-based experiences were nonsignificant. Implications for research and clinical practice are discussed.

Perspective: Partners' helping motivations and ICPs' psychological needs seem to be important to consider when investigating the role of spousal responses, because they could (indirectly) predict changes in the well-being and psychological distress of ICPs over time.

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Key words: Chronic pain couples, longitudinal study, well-being, helping motivation, self-determination theory.

Partners of individuals with chronic pain (ICP) may differ considerably in their helping responses (eg, providing emotional, instrumental, and informational support.³ Therefore, various attempts have been undertaken to categorize helping responses of close others in terms of their expected implications for ICPs' functioning (eg, Fordyce⁹). To fully understand whether partners' helping response is truly helpful and promotes ICPs' adjustment, the present research draws on self-determination theory (SDT^{7,34}), a broad theory on human motivation. Within SDT, it is argued that all people have 3 inherent psychological needs: the need for autonomy (ie, experiencing a sense of psychological freedom and choice), relatedness (ie, feeling close and

connected to others), and competence (ie, experiencing a sense of effectiveness and mastery). These needs are essential, because their satisfaction serves as a fundamental nutrient for individuals' psychological well-being and adjustment,⁷ more broadly, as well as the quality of one's romantic relationship, more specifically. In addition to considering the extent to which helping responses are supportive for the basic *psychological needs* of the person in pain,^{7,30} the theory emphasizes different *types of motivation* underlying partners' helping responses,^{16,17} because not all motivations are said to be equally need-conducive.

Specifically, 2 broad types of *motivation* are distinguished: autonomous motivation and controlled

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motivation.³⁹ When autonomously motivated, individuals help others because they derive inherent satisfaction from the act of helping or because they see the value of their helping behavior, either for themselves or for the recipient of help. In contrast, when controlled motivated, help is provided to avoid criticism from the help recipient or out of guilt feelings and pressured loyalty toward the recipient of help. In other words, controlled motivated help is experienced phenomenologically as a "should," whereas autonomously motivated help emanates more willingly from the person's interests and commitments. Available research indicates that both the help provider^{6,8,14,20,24,28,33} and the recipient of help⁴⁴ benefit more when the help originates from autonomous instead of controlled motivation. Recently, the motivations underlying pain-related social support have received initial attention. Specifically, a diary study charting the variability in helping motivation and adjustment on a daily basis indicated that on days when partners reported to provide help for autonomous reasons, ICPs experienced less relational conflict and having received more help, an effect that could be accounted for by improvements in ICPs' need-based functioning. In addition, daily autonomous helping motivation contributed indirectly, that is, through improvements in need-based functioning, to ICPs' positive and negative affects, their satisfaction with the received help, and perceived disability.¹⁷

The present study is the first to assess the temporal associations between partners' helping motivation and ICPs' functioning across a 6-month period, with ICPs' relationship-based need satisfaction and frustration considered as intervening variables. Specifically, a series of structural models tested this in a very conservative way by controlling for the initial levels of all variables in each model and for all within-time associations. The present study examined whether partners' autonomous helping motivation at the baseline assessment (T1) would predict changes in ICPs' relationship-based need satisfaction and frustration 3 months after baseline (T2), which, in turn, would predict changes in ICPs' well-being, psychological distress, and disability 6 months after baseline (T3). We expected that 1) partners' autonomous helping motivation would predict increases in ICPs' satisfaction and decreases in ICPs' frustration of relationship-based needs, and 2) ICPs' need satisfaction would predict enhanced well-being and diminished psychological distress and disability, whereas opposite findings were expected for ICPs' need frustration.

Methods

Study Design and Procedure

The present study is part of a larger study, the "Helping Motivation Diary and Longitudinal study" (HMDAL study) among ICPs and their partner. In addition to the longitudinal study reported in the present study, the HMDAL study comprises diary assessments (see Kindt et al^{15,17}). Participants who gave their agreement to be informed about studies performed at our laboratory were contacted by telephone 1) to receive more information about the

present study and 2) to assess the inclusion criteria. If both partners in a couple reported to have chronic pain, the individual with the longest pain duration was chosen as the ICP. ICPs and their partner were asked to complete questionnaires at 3 time points, spread across 6 months. Assessment with a 3-month interval was chosen because 3 months are not only enough to observe changes in our key study variables¹⁷ but also offered us the practical advantage of carrying out the research within 6 months. The informed consents and baseline questionnaires (T1) were administered via a home visit. Both partners received a link and a personal code for completing the other questionnaires (T2 and T3) online on a survey tool called LimeSurvey. When no computer or internet was available or when participants did not have enough experience with computer or the internet, they received a paper version. At T1 (total N = 140 ICPs and 140 partners), 39 (27.86%) ICPs and 31 (22.14%) partners chose to use the paper versions of the questionnaires. As a sign of appreciation, all couples received a fee of 30 euros after completing the questionnaires at T2 and T3. To enhance the completion rates, we reminded the participants by means of e-mail, telephone, or both. This study was approved by the ethical committee of the Faculty of Psychology and Educational Sciences of Ghent University.

Participants

Participants were couples, recruited through the Flemish Pain League (FPL), an umbrella organization for ICPs, and the Flemish League for Fibromyalgia Patients (FLFP), an organization specifically oriented to individuals with fibromyalgia. Of all the invited couples, 79.21% agreed to take part (for more recruitment details, see Kindt et al^{15,17}); that is, 141 couples participated in the longitudinal study, of which 97 couples were members of the FPL and 44 couples were members of the FLFP. The inclusion criteria for the participation of ICPs in the present study were 1) having chronic pain for at least 3 months, 2) living physically with a partner for at least 1 year, and (3) being sufficiently proficient in Dutch.

Although 141 couples agreed to take part in the "HMDAL study," data of only 140 ICPs and 140 partners were collected (the data of 1 ICP got lost via regular mail, and 1 partner was not at home during the home visit of T1). For T1, complete data were available for 139 couples. At T2 (3 months later than T1), 134 partners and 134 ICPs participated again. At T3 (6 months later than T1), 131 partners and 129 ICPs participated. For the current study, cases with missing values were included in the analyses, using the full-information maximum likelihood estimation.²¹ The full-information maximum likelihood estimation can be used only when no systematic differences are found between participants who stay in and participants who drop out; therefore, both subsamples were compared on the study variables. With regard to the ICP, Little's²¹ missing completely at random test produced a normed χ^2 (χ^2/df) of 112.25/100, $P = .19$, which means that no significant differences were found between the ICPs who participated 3 times and those ICPs who participated only once or twice. With regard to the partners,

Little's²¹ missing completely at random test was equally found to be nonsignificant, with χ^2 (χ^2/df) of 3.58/5, $P = .61$, thus implying that no significant differences were found between partners who participated 3 times and those partners who participated only once or twice. This indicates that the data were likely to be missing at random and that it is safe to impute cases with missing values in the analyses.

At T1, most of the ICPs ($n = 115$, 82.1%) were female; the mean age of ICPs and their partners ($n = 115$; 80.7% males) was 52.38 years (standard deviation [SD] = 11.72) and 53.61 years (SD = 12.02), respectively. All couples were heterosexual (except for 2) and white. More than a third (36.9% of ICPs and 34.3% of partners) of the sample reported that they have followed education beyond the age of 18 years. Almost all couples (81.5%) were married or legally cohabiting, with the mean relationship duration being 25.22 years (SD = 14.96). Most of the partners ($n = 91$, 65%) were employed, while 29 (20.7%) of the ICPs were employed. Almost all ICPs reported pain in > 1 location ($M = 4.02$, $SD = 1.68$, range = 1–7), with pain in the back (90.1%), neck (75.2%), and lower extremities (62.1%) being reported most frequently. Mean pain duration was 15.49 years (SD = 13.15). On a scale from 0 to 10, ICPs reported a mean pain intensity of 6.90 (SD = 1.41) and a mean disability of 6.50 (SD = 1.94). Fifty-six (40.3%) partners also reported pain complaints during the past 3 months (which is similar to other studies with couples with chronic pain, eg, Issner et al).¹³ Paired-sample t -tests showed that pain duration ($M = 9.84$, $SD = 11.87$), pain intensity ($M = 4.30$, $SD = 1.72$), and disability ($M = 2.77$, $SD = 2.21$) were significantly lower in partners compared with the ICPs (all P values < .05).

Measures

Helping Motivation

At all measurement times (T1, T2, and T3), partners' helping motivation was assessed by the Motivation to Help Scale^{18,44} that was modified in a previous study for use with couples with chronic pain (see Kindt et al¹⁶). Specifically, partners received a list of 20 reasons (instead of the original set of 11 items) for helping or supporting their partners in pain. Partners reported how true these motivations for helping were for them on a 7-point scale ranging from "1" (not at all true) to "7" (totally true). Drawing from SDT, 4 different types of motivation were distinguished: external motivation (5 items, eg, "because my partner would criticize me"), introjected motivation (5 items, eg, "because I would feel guilty if I didn't help"), identified motivation (5 items, eg, "because I think it is important to help my partner"), and intrinsic motivation (5 items, eg, "because I enjoy helping my partner"). In line with theory⁷ and previous research,^{16,17} items of the external and introjected motivation were summed up to represent controlled motivation to help, whereas items of the identified and intrinsic motivation were summed up to represent autonomous motivation to help. Cronbach

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α for T1, T2, and T3 were .88, .87, .90, respectively, for autonomous helping motivation and .78, .81, .80, respectively, for controlled motivation. In line with previous work (eg, Kindt et al¹⁷ and Weinstein and Ryan⁴⁴), an overall index reflecting the relative autonomous helping motivation (RAHM) was calculated by subtracting the controlled motivation scores from the autonomous motivation scores.

Relationship-Based Need Satisfaction and Frustration

The need satisfaction and frustration of the ICPs were assessed at each time point with the Basic Psychological Need Satisfaction and Frustration Scale,² adapted for use within intimate relationships (see Vanhee et al^{37,38}). The 24 items were scored on a 5-point Likert scale ranging from "1" (completely untrue) to "5" (completely true). Higher scores reflect higher levels of need satisfaction or frustration. All items started with "In the relationship with my partner" and were, for example, "..., I can freely take decisions" (ie, autonomy satisfaction), "..., I am confident that I can do things right" (ie, competence satisfaction), "..., I feel that s/he cares about me" (ie, relatedness satisfaction), "..., most of the things I do feel like I have to" (ie, autonomy frustration), "..., I have serious doubts about whether I can do things well" (ie, competence frustration), and "..., I sometimes have the impression that s/he dislikes me" (ie, relatedness frustration). Participants' relationship-based need satisfaction and frustration scores were computed by summing up scores for all items included in each of the 3 subscales for satisfaction or frustration. Need satisfaction showed good reliability (T1 $\alpha = .84$, T2 $\alpha = .85$, and T3 $\alpha = .88$), and need frustration had a good internal consistency (T1 $\alpha = .85$, T2 $\alpha = .88$, and T3 $\alpha = .90$).

Well-being

To measure subjective "well-being," we tapped into 2 different aspects: a cognitive evaluation of one's quality of life (QoL) and one's experience of positive emotions. The overall QoL in ICPs was measured every time by using a linear analog scale.²⁵ This is a vertical-graded, 10-cm line, ranging from "0" (the worst imaginable QoL) to "100" (the best imaginable QoL). The use of this rating scale allows ICPs to give a rating of their overall perceived QoL. To measure the positive affect, the positive affect subscale (10 items, eg, enthusiastic) of the widely used Positive and Negative Affect Schedule⁴³ was administered. Individuals with chronic pain completed this questionnaire at each time point. Each item was rated on a 5-point scale ranging from "1" (very slightly) to "5" (extremely) to indicate the extent to which the affect was experienced during the past 2 weeks. Cronbach α in the current study were .88, .90, and .88 for T1, T2, and T3, respectively. For the present study, scores for "well-being" were computed by averaging the standardized scores for the overall QoL and positive affect.

Psychological Distress

To measure ICPs' "psychological distress," we used the subscales of the Dutch 21-item version⁵ of the Depression Anxiety Stress Scale (DASS),²² which is designed to measure the negative emotional states of depression, anxiety, and stress during the past week. Each of the 3 DASS scales contains 7 statements that are to be rated on a 4-point Likert Scale ranging from "0" (not at all) to "3" (very much); for example, "I was unable to become enthusiastic about anything" (depression), "I felt scared without any good reason" (anxiety), and "I found it difficult to relax" (stress). Cronbach α in the current study were .88, .90, and .91 for depression, .80, .82, and .85 for anxiety, and .88, .91, and .91 for stress for T1, T2, and T3, respectively. In the present study, scores from the 3 DASS subscales were used as indicators for our latent variable "psychological distress"

Pain Intensity and Disability

Pain intensity of ICPs at T1 and disability (assessed at all 3 time points) were assessed with the Graded Chronic Pain Scale.⁴² A pain intensity score was calculated by averaging 3 ratings for pain intensity (current pain, average pain, and worst pain in the past 6 months), each on a scale ranging from "0" (no pain) to "10" (worst imaginable pain). In the present study, Cronbach α was .76. A disability score was computed by calculating the mean score of 3 items assessing the interference of pain with activities (daily activities; recreational, social, and family activities; work or household activities) during the past 3 months, which were also rated on a scale ranging from "0" (no interference) to "10" (impossible to carry out activity). Cronbach α were .88, .92, and .92 for T1, T2, and T3, respectively.

Statistical Plan

First, to explore the data, correlational analyses were conducted, and the mean and SD of the key study variables were inspected. Next, preliminary analyses were conducted to examine the differences in the study variables in terms of ICP's sex, chronic pain in both versus a single partner, age (both partners and the ICP), relationship duration, ICP's pain duration, and ICP's pain intensity. Based on these preliminary analyses, a choice was made as to which control variables were further accounted for in the primary analyses. The primary analyses concerned structural equation analyses, as tested in MPlus 7.4.²⁶ As suggested by Hu and Bentler (1999),¹² we evaluated model fit based on a combined consideration of the χ^2 statistic, the comparative fit index (CFI), the root mean square error of approximation (RMSEA), and the standardized root mean square residual (SRMR). The χ^2 should be as small as possible. A CFI value of .90 or higher indicates a reasonable fit, whereas for RMSEA and SRMR, values $\leq .08$ indicate acceptable fit.¹⁹

Results

Descriptive Statistics and Preliminary Analyses

Means, SD, and correlations among the study variables are presented in Table 1.

A multivariate analysis of covariance examined the effects of several important background variables (ie, ICP's sex, the presence of chronic pain in both versus a single partner, partner and ICP's age, relationship duration, ICP's pain duration, and ICP's pain intensity) on all our study variables measured at T1, T2, and T3. A significant multivariate effect for ICPs' pain intensity (Wilk's $\lambda = .39$; $F(18, 94) = 8.23$, $P < .001$) was obtained. Subsequent univariate effects were present for ICPs' need satisfaction at T2 ($F(1, 111) = 6.72$, $P < .05$), with higher pain intensity linked to less need satisfaction ($r = -.19$, $P < .05$). Univariate effects for ICPs' pain intensity were also significant for ICPs' well-being at T2 ($F(1, 111) = 27.33$, $P < .001$) and T3 ($F(1, 111) = 4.05$, $P < .05$), for ICPs' psychological distress at T1 ($F(1, 111) = 11.15$, $P < .01$), T2 ($F(1, 111) = 13.47$, $P < .001$), and T3 ($F(1, 111) = 6.62$, $P < .05$), and for ICPs' disability at T1 ($F(1, 116) = 58.47$, $P < .001$), T2 ($F(1, 111) = 26.94$, $P < .001$), and T3 ($F(1, 111) = 29.44$, $P < .001$). The direction of effects was similar in each of these cases (see Table 1), with ICPs with higher pain intensity at T1 reporting lower well-being, higher psychological distress, and higher disability. Based on this multivariate analysis of covariance, we decided to control for ICPs' pain intensity in all subsequent analyses.

Structural Equation Modeling

Well-being

Regarding ICPs' well-being, we tested 2 models, which are graphically displayed in Fig 1. The first model (model 1a) was tested with ICPs' relationship-based need satisfaction as an intervening variable in the relation between partners' helping motivation and ICP well-being, and the second model (model 1b) was tested with ICPs' relationship-based need frustration as an intervening variable. Estimation of model 1a ($\chi^2(25) = 39.37$, $P = .03$, RMSEA = .06, CFI = 0.98, SRMR = .06) showed no significant effect of partners' helping motivation on ICPs' relationship-based need satisfaction across time. However, relationship-based need satisfaction was related to ICP well-being across time ($\beta = .14$, $P < .05$ from T2 to T3). Replacing need satisfaction by need frustration in model 1b ($\chi^2(24) = 30.60$, $P = .17$, RMSEA = .04, CFI = .99, SRMR = .04) indicated that partners' autonomous helping motivation predicted significant decreases in ICPs' need frustration across time ($\beta = -.15$, $P < .01$ from T1 to T2). Further, from T1 to T2, ICPs' need frustration and ICPs' well-being were negatively related, with bidirectional relationships emerging.

Psychological Distress

For "psychological distress," we followed a 2-step procedure,¹ because the 3 DASS subscales were used as indicators

Table 1. Pearson Correlations, Means, and SDs Among the Measured Variables

	T1					T2					T3									
	RAHM	NS	NFR	WB	PD	DIS	RAHM	NS	NFR	WB	PD	DIS	RAHM	NS	NFR	WB	PD	DIS	PAIN INTENSITY	
RAHM	—	.31***	-.27**	-.00	-.18*	-.01	.72***	.28**	-.29**	.12	-.20*	-.04	.73***	.23**	-.25**	.08	-.18*	.08	-.05	
NS		—	-.66***	.31***	-.39***	-.10	.24**	.71***	-.66***	.23**	-.36***	-.11	.19*	.75***	-.57***	.25**	-.40***	-.04	-.10	
NFR			—	-.22**	.46***	.21*	-.20*	-.57***	.71***	-.26**	.45***	.15†	-.21*	-.54***	.68***	-.10	.47***	.07	.18*	
WB				—	-.44***	-.43***	.00	.29**	-.29**	.61***	-.35***	-.29***	-.03	.21*	-.25**	.43***	-.40***	-.22*	-.30***	
PD					—	.35***	-.14	-.32***	.39***	-.43***	.84***	.20*	-.13	-.30**	.36***	-.31***	.81***	.22*	.32***	
DIS						—	.02	-.08	.11	-.48***	.30***	.68***	.02	-.05	.09	-.25**	.24**	.64***	.61***	
RAHM							—	.25**	-.31***	.14	-.16†	-.08	.78***	.25**	-.31***	.10	-.19*	-.03	.09	
NS								—	-.73***	.33***	-.37***	-.10	.15†	.76***	-.66***	.29***	-.43***	.03	-.19*	
NFR									—	-.22*	.47***	.14	-.26**	-.74***	.83***	-.18*	.51***	.01	.05	
WB										—	-.44***	-.57***	.10	.23**	-.27**	.55***	-.43***	-.41***	-.45***	
PD											—	.29**	-.19*	-.31***	.45***	-.30***	.83***	.23**	.33***	
DIS												—	-.12	-.11	.20*	-.27**	.24**	.64***	.44***	
RAHM													—	.17†	-.24**	.06	-.22*	-.02	.13	
NS														—	-.69***	.26**	-.42***	.04	.04	
NFR															—	-.24**	.57***	.05	.14	
WB																—	-.40***	-.34***	-.23*	
PD																	—	.23**	.26**	
DIS																		—	.46***	
Mean	19.75	45.25	26.36	.04	-.00	6.50	19.74	45.37	25.46	.00	-.00	6.48	19.52	44.84	25.68	-.03	-.00	6.27	6.90	
SD	12.77	7.13	8.20	12.90	7.07	1.94	13.82	7.39	8.82	12.54	7.79	2.02	13.89	8.10	9.36	11.94	8.13	2.13	1.41	

Abbreviations: RAHM, partners' relative autonomous helping motivation; NS, ICPs' relationship-based need satisfaction; NFR, ICPs' relationship-based need frustration; WB, ICPs' well-being; PD, ICPs' psychological distress; DIS, ICPs' disability.

† $P < .10$.

* $P < .05$.

** $P < .01$.

*** $P < .001$.

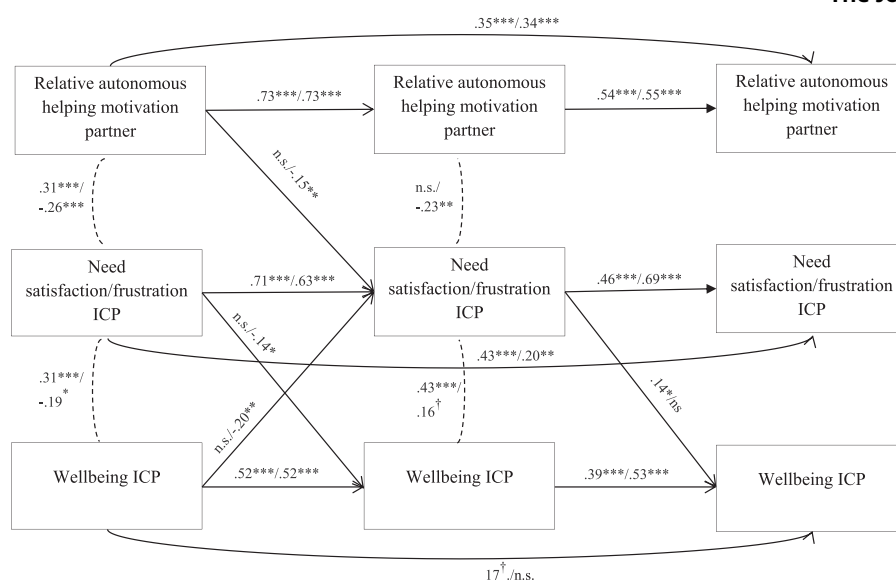


Figure 1. Structural equation model of partners' helping motivation, ICPs' need-based experience, and ICPs' well-being. Note that path coefficients of model 1a with need satisfaction are presented before brackets, whereas path coefficients of model 1b with need frustration are presented after brackets. We controlled for ICP pain intensity reported at T1. Coefficients shown are standardized path coefficients: † $P < .10$, * $P < .05$, ** $P < .01$, *** $P < .001$. For the sake of parsimony, only significant associations are presented.

for our latent variable. First, a confirmatory factor analysis was used to test the quality of the measurement model of the study constructs. Second, the structural models were tested. The measurement model concerning psychological distress included 3 latent variables (ie, psychological distress, as measured at 3 measurement points) and 9 indicators (ie, depression, anxiety, and stress as 3 indicators of psychological distress, each measured at 3 measurement points). The measurement errors of the same indicators across time were allowed to covary, because the same indicators share error variance over different moments in time.³² Further, the factor loadings of the same indicators were set equal across the 3 measurement points. The model fitted the data adequately ($\chi^2(19) = 16.73$, $P = .61$, CFI = 1.00, RMSEA = .00, SRMR = .03). Moreover, all factor loadings were highly significant ($P \leq .001$), with values ranging from .82 to .88 ($M = .85$).

Next, we tested 2 models, which are graphically displayed in Fig 2, with ICPs' psychological distress as the outcome, where the first model (ie, model 2a) considered ICPs' relationship-based need satisfaction as a potential intervening variable and the second model (ie, model 2b) considered ICPs' relationship-based need frustration as a potential intervening variable. Estimation of model 2a ($\chi^2(84) = 100$, $P = .11$, RMSEA = .04, CFI = .99, SRMR = .05) showed no significant effect of partners' helping motivation on ICPs' need satisfaction. However, ICPs' need satisfaction significantly predicted a decrease in psychological distress over time ($\beta = -.21$, $P < .001$). Estimation of model 2b ($\chi^2(3) = 93.99$, $P = .19$, RMSEA = .03, CFI = .99, SRMR = .05) showed a significant effect of partners' autonomous helping motivation on decreases in ICPs' need frustration from T1 to T2 ($\beta = -.11$, $P < .05$). Further, ICPs' need

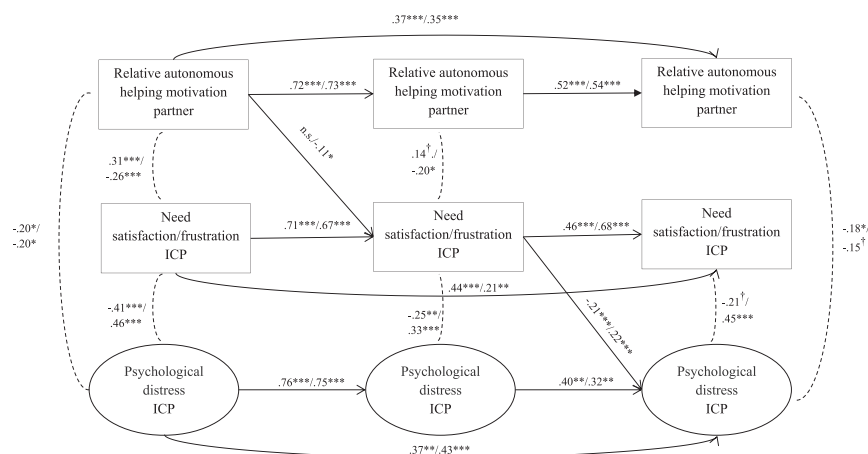


Figure 2. Structural equation model of partners' helping motivation, ICPs' need-based experience, and ICPs' psychological distress. Note that path coefficients of model 2a with need satisfaction are presented before brackets, whereas path coefficients of model 2b with need frustration are presented after brackets. We controlled for ICP pain intensity reported at T1. Coefficients shown are standardized path coefficients: † $P < .10$, * $P < .05$, ** $P < .01$, *** $P < .001$. For the sake of parsimony, only significant associations are presented.

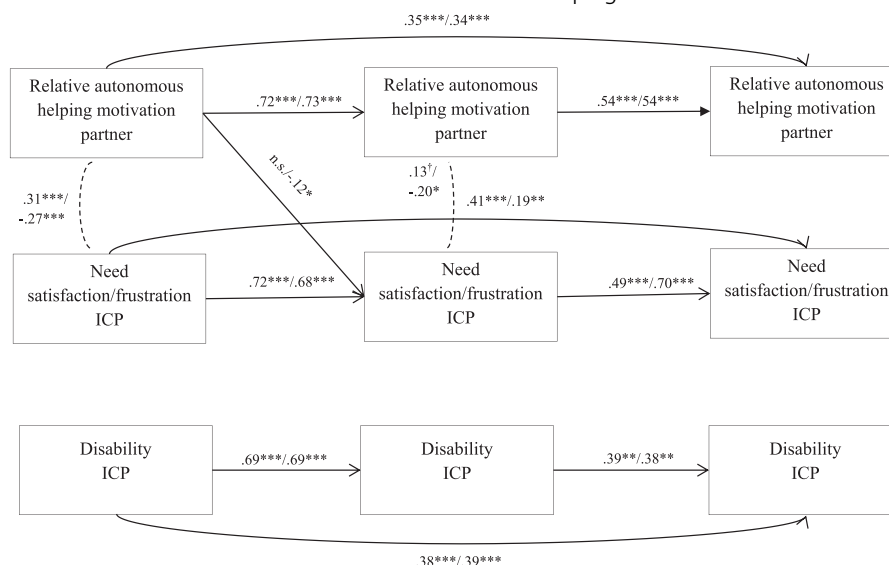


Figure 3. Structural equation model of partners' helping motivation and ICPs' need-based experience, and ICPs' disability. Note that path coefficients of model 3a with need satisfaction are presented before brackets, whereas path coefficients of model 3b with need frustration are presented after brackets. We controlled for ICP pain intensity reported at T1. Coefficients shown are standardized path coefficients: † $P < .10$, * $P < .05$, ** $P < .01$, *** $P < .001$. For the sake of parsimony, only significant associations are presented.

frustration was significantly related to an increase in ICPs' psychological distress ($\beta = .22$, $P < .001$). When testing whether the indirect path from partners' helping motivation to ICPs' psychological distress via ICPs' relationship-based need frustration was significant,²⁹ the results showed that this was not the case ($\beta = -.02$, $P = .11$). This indirect effect may not have reached full significance owing to the conservative way in which paths were tested (ie, controlling for the initial levels of all variables in the model and for within-time associations) and also given the multi-informant methodology.

Disability

Fig 3 provides an overview of the results for ICPs' disability, where the first model (ie, model 3a) tested ICPs' relationship-based need satisfaction as an intervening variable and a second model (ie, model 3b) tested ICPs' relationship-based need frustration as an intervening variable. As for model 3a ($\chi^2(28) = 46.34$, $P = .02$, RMSEA = .07, CFI = .97, SRMR = .05), there was no significant effect of partners' helping motivation on ICPs' need satisfaction and no significant effect of ICPs' need satisfaction on disability. Model 3b ($\chi^2(27) = 39.93$, $P = 0.05$, RMSEA = .06, CFI = .98, SRMR = .04), involving need frustration as an intervening variable, indicated that, as noted previously, partners' autonomous helping motivation predicted a decrease in ICPs' need frustration across time ($\beta = -.12$, $P < .05$), but ICPs' need frustration failed to relate to disability across time.

Discussion

In this study, we investigated whether partners' helping motivation would indirectly relate to ICP outcomes, via ICPs' need-based experiences, as defined by SDT.³⁴

Need satisfaction is distinguished from frustration, because, by definition, the absence of satisfaction does not denote its frustration. Partners can either support or thwart each other's needs; a lack of need satisfaction involves being indifferent toward the partner's needs, whereas need frustration involves a more active way of undermining the partner's needs.⁴⁰

The Role of ICPs' Need-Based Experiences

Results showed an effect of partners' helping motivation on ICPs' basic psychological needs, which in turn had an effect on ICPs' psychological functioning. However, the indirect effect did not reach significance. Specifically, when partners provide help volitionally instead of feeling pressured to do so (ie, autonomous, relative to controlled, helping motivation), ICPs report less relationship-based need frustration over time, whereas the association with ICPs' need satisfaction was nonsignificant, which was rather surprising, considering the significant correlations. The multi-informant nature of the data, involving separate reports of partners and ICPs, and the conservative way of testing may be possible reasons for this, as are the high mean scores for need satisfaction, suggesting that there was not much room for improvement in need satisfaction over time. With regard to ICPs' need frustration, results were in line with a diary study in which the fluctuations in partners' daily helping motivations were predictive of the changes in ICPs' daily functioning.¹⁷

When focusing on the associations between ICPs' need-based experiences (ie, the need for autonomy, competence, and relatedness within a romantic relationship) and ICP outcomes, the present results partially supported the proposition that basic psychological needs are essential ingredients for optimal functioning.¹¹ More specifically, need satisfaction

predicted an increase in ICPs' well-being and a decrease in ICPs' psychological distress, whereas need frustration predicted an increase in ICPs' psychological distress. This finding is in line with the main postulates of SDT about the importance of 3 basic psychological needs for psychological well-being⁷ and with another study showing that ICPs' need-based experiences were predictive of ICPs' daily functioning,¹⁷ self-esteem, life satisfaction, and psychological symptoms 6 months later.³⁶

With regard to ICPs' disability over time, no significant associations were found either with ICPs' need-based experiences or with partners' helping motivation. This is not in line with a previous diary study,¹⁷ in which partners' daily autonomous helping motivations related indirectly, through ICPs' relationship-based need satisfaction and frustration, to the change in ICPs' daily disability. One possible hypothesis is that the need-based experiences play a role only in the short-term within-person differences in disability, whereas it is more difficult to influence disability at the between-person level.

Theoretical and Clinical Implications

A romantic partner is especially impactful in a person's life. Romantic partners often have the ability to pressure and persuade the other partner to adhere to medical treatment, encourage well behaviors (eg, exercise), and provide emotional support, potentially leading to a faster recovery.³⁵ The present study provided additional evidence for the important role that spousal responses play in ICPs' well-being. Specifically, findings suggest that partners' underlying motivation for help is important to consider when investigating the role of different spousal responses, independently of the specific kind of behavior. In this study, we found evidence for the temporal associations between partners' helping motivation and ICPs' need frustration. Autonomously motivated partners might be less rigid and more flexible in prioritizing ICPs' needs above their own needs and may be more receptive for the feedback of the ICP in the caregiving process.⁴¹ An autonomous helping motivation may prevent partners from becoming overprotective^{10,11} or solicitous^{4,31} and thereby buffer against thwarting ICPs' need for autonomy (eg, receiving unwanted or unnecessary help), competence (eg, feeling incapable of taking care for oneself), and relatedness (eg, cold interaction or feeling distance). These results reveal that it is important to provide a need-supportive environment to patients.

Limitations and Future Research

There are some limitations that should be considered. First, despite our conservatory analyses, causality cannot be inferred, as third, unmeasured variables may account for the observed associations. Future research should better unpack the direction of effects by using experimental designs priming partners' helping motivation. Second, the used

measures are all self-report scales, which may create a response bias through the phenomenon of social desirability. Third, most theoretically relevant path coefficients were small in size. Given the multi-informant nature of the data and the conservative way of testing this is not totally unexpected. However, future research with a larger sample of participants should aim to replicate the current pattern of findings before drawing firm conclusions. Fourth, the study sample mostly included female patients, with a long relationship duration, so caution is needed in generalizing the present study's results to diverse populations. Future research should include more male patients and patients in both short and long term relationships.

Further, as the current study had the explicit intention to define helping responses as broad as possible (consistent with previous research⁴⁴), it would be very interesting in the future research to specify the type of help being provided and to examine whether the motivation for 1 type differ from another type. Finally, as previous research has shown that helping responses that promote functional dependence are generally considered more maladaptive,^{23,27} a final issue that deserves further attention in the future research would be to investigate whether ICPs' functional independence gets promoted when ICPs' psychological needs are met. Of course, next to these limitations and directions for the future research, this study also has several strengths, for example, the low dropout of participants and the multi-informant approach. With regard to relationship duration, the fact that all couples lived together for at least 1 year could also be seen as a strength of the current study, because these participants may have developed a more stable pattern of motivational functioning that is less susceptible for change.

Conclusions

The current study shows that partners' autonomous, relative to controlled, helping motivation has have a positive effect on ICPs' relationship-based need frustration. Relationship-based need satisfaction and frustration of the ICPs were predictive of the changes in their well-being and psychological distress over time. Future research could further explore how partners can nurture ICPs' needs and identify other antecedents of partners' helping motivation (eg, goal conflict¹⁵), which could then be used as targets for clinical interventions.

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